

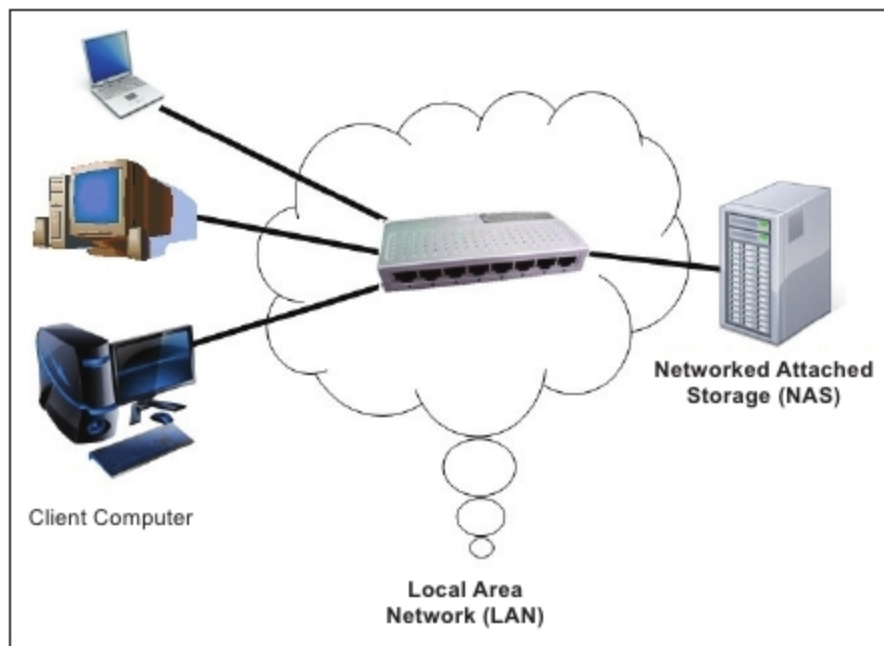


Figure 1: Network attached storage with a LAN network

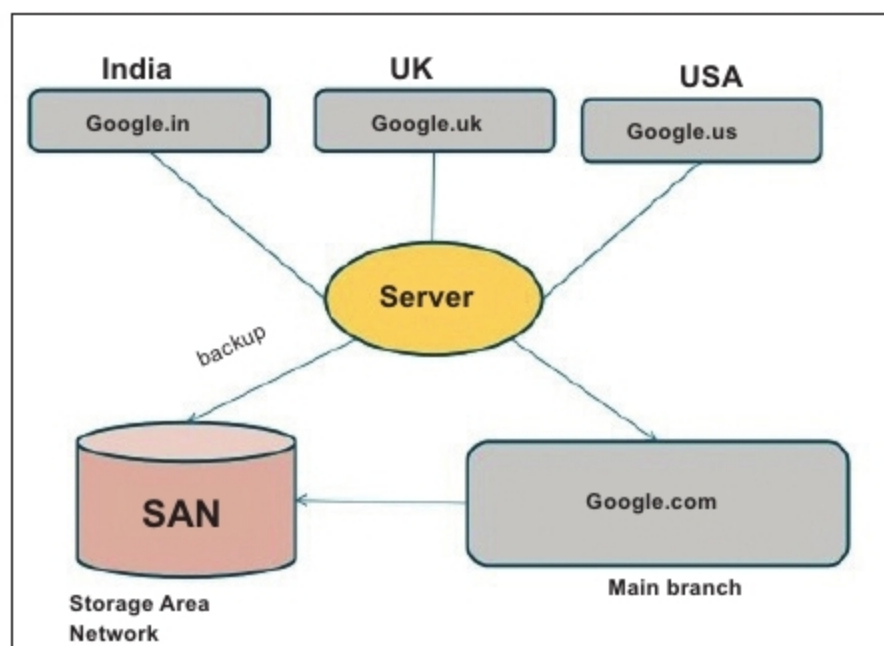


Figure 2: Storage area network with a LAN network

the case) or the network slacks.

Storage area network

Storage area network (SAN) is a rapid system of storage devices and switches associated with computer systems. SAN works as a result of sharing the pool with multiple servers. Each server can have direct access to the storage. SAN provides centralised storage management. It can move information starting with one storage device onto the next, and can share information between different servers, which makes the restoration and backup of information quick. A well designed SAN will be more productive and enables fast recuperation from calamities, as well as high accessibility.

Features of SAN: SAN uses block level data access. The main I/O protocol of SAN is the Small Computer System Interface (SCSI). SAN appears to the computer as its own storage. It is connected by means of the fibre channel, which results in excellent speed and performance.

Nowadays, many organisations and sectors have switched to software defined infrastructure which means they have to safely and securely store large amounts of data. Many of the organisations have been using open source software to meet

their requirements for storing data — for example, projects like Hadoop, Ceph, Gluster, etc.

Home users and small businesses can also use open source storage software to meet their needs of storing data. This software allows them to set up their own NAS or SAN devices, which use industry standard hardware, thus avoiding paying the high price for dedicated storage equipment. It is also possible to set up cloud storage which is safe, secured and private, and affordable too when organisations look for backup and recovery.

A few NAS and SAN open source software

Ceph: Red Hat's Ceph offers multi-protocol (i.e., unified) objects and block storage capabilities. It is a distributed storage solution that boasts of scalability, reliability and excellent performance. Well-known users include Cisco, DreamHost, CERN, Bloomberg and Deutsche Telekom.

Operating system: Linux

CryptoNAS: This project aims to simplify the process of setting up an encrypted file server. It comes in a live CD package or a server package that adds a Web front-end.

Operating system: Linux

ESOS: Short for Enterprise Storage OS, ESOS is a Linux distribution for setting up a storage array on your own hardware. Commercial support is available.

Operating system: Linux

FreeNAS: FreeNAS claims to be the world's no.1 storage OS with over 10 million downloads. It counts the United Nations, the University of Florida, the Salvation Army, Reuters, Michigan State University, the Department of Homeland Security and many other organisations among its users. It can be installed on nearly any hardware to turn it into a network attached storage (NAS) device. Paid, supported enterprise solutions based on the same technology are available under the TrueNAS brand name.

Operating system: FreeBSD

NAS4Free: NAS4Free claims to be "the simplest and fastest way to create a centralised and easily accessible server for all kinds of data." Its features are Zettabyte File System (ZFS), software RAID (levels 0, 1 or 5) and disk encryption.

Operating system: FreeBSD

Openfiler: As a unified storage solution, Openfiler combines the best features of NAS and SAN storage. Key features include high availability/failover, block replication and Web-based management. The London Metropolitan Police, Bill Me Later, Pratt & Whitney and Motorola are some of the users. A paid commercial version is also accessible.

Operating system: Linux

OpenMediaVault: OpenMediaVault is described on its site as a "next-generation network attached storage (NAS) solution." This system was mainly designed to provide NAS for home users as well as small offices. It offers Web based organisation support and incorporates programming RAID abilities.

Operating system: Linux

Turnkey Linux File Server: The Turnkey Linux Project